UNICERT: A DIGITAL DOCUMENT ISSUANCE AND VERIFICATION SYSTEM FOR UNIVERSITIES

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ABSTRACT

Universities continue to rely heavily on paper-based documents, handwritten signatures, physical stamps, and manual administrative procedures for a wide range of academic and administrative activities. Although digital certificates and electronic signatures provide established mechanisms for ensuring document integrity, authenticity, and signer identification, their practical adoption within university-wide administrative processes remains limited in many contexts.

This project proposes UniCert, an integrated digital document certification and verification system designed for the University of El Oued. The proposed system will combine a central document issuance and management system, dedicated digital signing software, and a public document verification portal. These components will work together to allow university documents to be created and approved within the institution, digitally signed by authorized personnel, securely stored, and later verified by third parties to confirm their authenticity and integrity. The system will be designed to support the generation, approval, signing, storage, and verification of a broad range of university documents rather than being restricted to a single document type such as diplomas.

The project will use the University of El Oued as its case study while maintaining a modular and scalable architecture that can be configured for other institutions. Particular emphasis will be placed on security, certificate-based trust, document integrity, traceability, accountability, and the automation of routine administrative activities. A working prototype of the system will be developed and evaluated under controlled conditions, with synthetic or appropriately controlled data where access to real institutional data cannot be obtained within the project timeframe.

1 KEYWORDS

Digital certificates, digital signatures, document verification, public key infrastructure, university information systems, electronic document management.

2 INTRODUCTION

Digital transformation has increasingly changed the way organisations create, process, store, exchange, and verify information. In particular, digital certificates and electronic signatures provide cryptographic mechanisms for establishing trust in electronic documents by linking documents to the identity of their signers and allowing recipients to determine whether the document has been tampered with or altered after signing. These mechanisms form an important part of modern electronic administration and trusted digital transactions.

In Algeria, the development of electronic certification and signatures has been supported by a legal and institutional framework. Law No. 15-04 of 1 February 2015 established the general rules relating to electronic signatures and electronic certification and provided the legal foundation for the national electronic certification framework [1]. The framework has since evolved. In February 2026, Law No. 26-02 established the general rules relating to trust services for electronic transactions and electronic identification and repealed Law No. 15-04, while maintaining its implementing provisions during the transition to the new framework [2]. The new law explicitly addresses electronic signatures, electronic seals, trust-service providers, certificate validity, and electronic identification.

The existence of such a framework, however, does not by itself provide an institution with an integrated system for applying digital certification to its everyday administrative activities. University administration represents a particularly relevant environment for such a system because universities produce a large and diverse range of documents involving students, teaching staff, administrative staff, departments, faculties, and central administration.

In many university environments, the digitisation of individual services does not necessarily eliminate the underlying dependence on paper documents. Documents may still need to be requested manually, processed by an administrative office, printed, signed or stamped physically, and subsequently collected by the recipient. Even where digitally signed documents are available, signing may depend on general-purpose third-party software rather than an institutional system designed around the university's own document generation, authorisation, and verification processes.

A more complete approach is to treat digital certification as an institutional infrastructure rather than as an isolated feature attached to one particular type of document. Such an infrastructure should allow authorised personnel to generate documents from controlled templates, apply the appropriate institutional signature or electronic seal, preserve a record of the issued document, and provide an independent mechanism through which a recipient or third party can verify the authenticity and integrity of the document.

This project proposes UniCert as such an infrastructure, using the University of El Oued as its case study. The system will consist of three principal components: a central document issuance and management system, dedicated digital signing software, and a public verification portal. These components will operate as parts of one integrated trust and document-management architecture.

The proposed system is not intended to be limited to diplomas or a single administrative process. Instead, it will be designed around configurable document types, institutional roles, approval workflows, certificate policies, and verification procedures. The University of El Oued therefore serves as the initial deployment context while the architecture is intended to remain sufficiently modular and scalable to support adaptation to other universities or organisations.

The project also addresses the broader objective of reducing unnecessary dependence on paper-based administrative processes. By automating document generation and approval workflows, applying verifiable digital signatures or seals, and providing persistent records and verification mechanisms, the system aims to contribute to a more sustainable "zero-paper" university environment while improving administrative traceability and accountability.

3 Related Work

Research and practical systems surrounding digital academic certification demonstrate that universities can use electronic signatures and digital certificates to replace or complement traditional document-processing procedures. However, existing implementations differ considerably in their scope, architecture, and degree of automation.

3.1 Algerian Legal and Institutional Context

The Algerian legal framework for electronic certification initially took a formal shape through Law No. 15-04 of 1 February 2015, which established the general rules relating to electronic signatures and electronic certification [1]. Academic and legal literature subsequently examined the implications of this framework, including the role of certification service providers in establishing trust in electronic documents [3].

The legal framework was substantially updated by Law No. 26-02 of 17 February 2026, which establishes general rules for trust services for electronic transactions and electronic identification [2]. Among other provisions, the law defines electronic signature certificates and electronic seals and establishes requirements concerning the validity and verification of qualified certificates. It also introduces explicit provisions concerning electronic identification and different levels of assurance.

These developments demonstrate that the legal foundations for trusted electronic transactions exist and continue to evolve. The present project therefore focuses not on establishing the legal validity of

electronic signatures, but on investigating how these mechanisms can be incorporated into a practical university-wide information system.

3.2 University Politécnica de Madrid

The University Politécnica de Madrid (UPM) provides a useful practical reference for university electronic administration. Its electronic seat supports procedures through which students can request academic certificates electronically and subsequently download certificates signed using an electronic seal belonging to the university [4].

UPM also maintains electronic seals associated with different administrative functions. Its public documentation identifies electronic seals for procedures including electronic administration, academic management, continuing education, research management, and the General Secretariat [5]. This demonstrates the value of separating institutional signing identities according to administrative roles and functions.

The UPM electronic seat also provides a public document-validation service for signed PDF documents and provides mechanisms for validating certificates used by the electronic seat [6]. The university further requires the use of recognised digital certificates and, where appropriate, the AutoFirma application for electronic procedures [7].

These features provide an important conceptual reference for UniCert: university-issued documents can be generated through electronic procedures, signed using institutionally managed certificates or seals, and subsequently verified independently. UniCert aims to investigate a similar model within the context of an Algerian university while placing particular emphasis on a modular architecture that can be adapted beyond a single administrative service.

Related work is also available in the UPM institutional repository. For example, a UPM undergraduate project proposes a cloud-based document management and signing solution incorporating blockchain technology and examines the relevant electronic-signature regulations and system requirements [8].

3.3 Identified Gap

The reviewed material demonstrates that digital certification of university documents is technically feasible and that university electronic administration systems can incorporate digital signatures, electronic seals, and document verification. However, the preliminary investigation suggests that the Algerian context lacks a publicly demonstrated university-wide implementation combining these functions into one coherent system for a broad range of ordinary academic and administrative documents.

The gap addressed by this project is therefore not the invention of digital signatures or digital certificates themselves. Rather, it concerns the design and implementation of an integrated institutional system that connects document generation, authorisation, signing, storage, traceability, and independent verification into a single workflow.

The project will investigate this gap through the development of UniCert, using the University of El Oued as the case study and designing the resulting system so that its principal components can be configured and reused by other institutions.

4 Type of Project

This project is primarily a development project with an applied research and evaluation component.

The development component consists of the design and implementation of the UniCert software system and its three principal components: the document issuance and management system, the digital signing software, and the public verification portal.

The applied research component consists of investigating existing digital certification practices, relevant legal and technical requirements, university certification systems, security and trust requirements, and the limitations of existing approaches in order to inform the proposed architecture.

The evaluation component will assess whether the resulting prototype provides the required security, integrity, traceability, verification, and workflow properties under controlled test conditions.

5 Aims and Objectives

5.1 Aim

The primary aim of this project is to design and implement a modular and scalable digital certification system for the University of El Oued that provides a secure institutional framework for generating, approving, digitally signing, issuing, storing, and independently verifying university documents.

5.2 Objectives

The project will pursue the following objectives, ordered according to their importance to the core purpose of the system:

1. Design and implement a strong security and trust model for university digital documents, including appropriate use of digital certificates, digital signatures or electronic seals, certificate validation, key management, access control, and document-integrity mechanisms.
2. Design and implement an integrated architecture connecting document generation and management, institutional digital signing, document storage, and independent public verification while maintaining clear separation of responsibilities between system components.
3. Provide reliable mechanisms for establishing the authenticity and integrity of issued university documents, allowing recipients and authorised third parties to independently verify documents without relying solely on the issuing office.
4. Improve administrative traceability and accountability by maintaining appropriate records of document creation, approval, signing, issuance, modification, and verification activities.
5. Automate recurring university administrative and document-processing activities, particularly document generation, approval workflows, signing, issuance, and verification, thereby reducing unnecessary manual intervention.
6. Address weaknesses and practical limitations identified in existing approaches to digital document certification in the Algerian context, while incorporating appropriate security and trust mechanisms to avoid reproducing those limitations.
7. Establish a foundation for a sustainable "zero-paper" university environment by replacing paper-dependent document workflows with digitally generated, signed, stored, and verifiable documents wherever institutional and legal requirements permit.
8. Develop the system using a modular and scalable design so that document types, organisational structures, administrative roles, certification policies, and verification procedures can be configured for other universities or organisations without redesigning the entire system.
9. Provide an institutional foundation that can support the University of El Oued's long-term digital transformation objectives and strengthen its readiness for international benchmarking and digitalisation initiatives, including indicators considered by international university-ranking frameworks such as QS and THE.
10. Demonstrate a realistic implementation that can position the University of El Oued as an early Algerian adopter of an integrated university digital certification infrastructure and provide a foundation for potential institutional deployment.

6 Expected Outcomes and Deliverables

The principal outcome of the project will be a working prototype of the UniCert system demonstrating an end-to-end workflow for the generation, authorisation, signing, issuance, storage, and verification of university documents.

The project will produce the following deliverables:

1. **Technical project report.** A comprehensive technical report documenting the problem domain, requirements, legal and technical context, security and trust model, system architecture, design decisions, implementation, testing, evaluation, limitations, and recommendations for future development.
2. **Working UniCert prototype.** A functional demonstration of the proposed system consisting of:
   - a central document issuance and management system;
   - dedicated digital signing software; and
   - a public document and signature verification portal.
3. **Controlled demonstration environment.** The prototype will be demonstrated using controlled, synthetic, or otherwise appropriately authorised data where access to real university records cannot be obtained within the available project timeframe. The demonstration will nevertheless reproduce the relevant operational workflows and security properties of the proposed system.
4. **Open-source implementation repository.** The source code of the developed software components will be provided in an organised repository, subject to any institutional, security, or licensing restrictions that may apply.
5. **System and deployment documentation.** Documentation describing the architecture, configuration, deployment requirements, certificate and key-management assumptions, and procedures required to reproduce the demonstration environment.

7 Methods

The project will combine software engineering, applied research, security analysis, and controlled system evaluation.

First, a requirements analysis will be conducted to identify the functional, security, trust, administrative, and verification requirements of a university digital certification system. Particular attention will be given to the roles of document creators, approvers, signers, certificate authorities, document recipients, and relying parties.

Second, a review of the Algerian legal and institutional framework and relevant international and academic work will be conducted. This will include the Algerian legislation governing electronic signatures and trust services, existing university digital certification systems, and research addressing secure electronic document management.

Third, the security architecture will be designed using established cryptographic and information-security principles. The design will consider certificate hierarchies, trust relationships, key protection, authentication, authorisation, certificate validation and revocation, document integrity, auditability, and separation of administrative responsibilities.

Fourth, the system will be developed iteratively. The implementation will be divided into the principal system components while maintaining clearly defined interfaces between them. This approach will allow the document management, signing, and verification components to be developed and tested independently while remaining part of the same institutional architecture.

Fifth, the resulting system will be evaluated through controlled test cases. The evaluation will examine document generation, signing, modification detection, certificate validation, verification, access control, audit records, and failure conditions. Where real institutional data or production credentials cannot be obtained, synthetic or controlled data will be used.

Finally, the results will be analysed against the original requirements and objectives. Limitations identified during implementation and evaluation will be documented, together with recommendations required for a future institutional deployment.

8 Resource Requirements

The project requires a number of technical and institutional resources.

8.1 Deployment Infrastructure

A dedicated server will be required to host the backend services and databases used by the prototype. For a formal deployment, the preferred configuration would be a physically hosted server under the control of the University of El Oued or an appropriately secured university infrastructure.

During development and demonstration, a local or virtualised environment may be used where access to university infrastructure is unavailable.

8.2 Domain and Network Infrastructure

A university-associated domain name using the `.dz` namespace will preferably be used for the public verification service and associated services. A production deployment will additionally require properly configured HTTPS, DNS, certificate management, firewalling, and other network-security controls.

8.3 Cryptographic Infrastructure

The prototype will require appropriate cryptographic key pairs and certificates for demonstrating the signing and verification workflows.

For a formal institutional deployment, protection of high-value private keys, particularly a root certification authority private key, would require dedicated hardware security infrastructure such as a Hardware Security Module (HSM) or an equivalent appropriately secured mechanism. This is considered an additional deployment requirement rather than a mandatory requirement for the prototype.

8.4 Development Software

The project will primarily rely on open-source development tools, cryptographic libraries, database systems, operating systems, and other software components wherever appropriate. The final technical report will document the selected technologies and their roles within the architecture.

8.5 Institutional Access and Test Data

Development and realistic evaluation may benefit from access to university document templates, organisational information, administrative workflows, and representative datasets. Where such access cannot be obtained within the project timeframe, controlled synthetic data will be used to reproduce the required workflows without exposing real personal or institutional information.

8.6 Literature and Documentation

The project will require access to academic publications concerning digital certificates, electronic signatures, public key infrastructure, secure document management, university information systems, and digital transformation, together with relevant Algerian legislation and institutional technical documentation.

9 Conclusion

UniCert proposes an integrated approach to university digital certification in which document generation, institutional signing, document management, and independent verification are treated as components of a single trusted system.

The University of El Oued will serve as the principal case study, but the system will be designed with modularity and scalability as fundamental requirements. The intended result is not simply a mechanism for producing digitally signed PDF files, but an institutional infrastructure capable of supporting a broader transition from paper-dependent administrative processes toward digitally generated, securely signed, traceable, and independently verifiable university documents.

The project will therefore provide both a practical software artefact and a technical investigation into how digital certification mechanisms can be adapted to the operational requirements of a university in the Algerian context.

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